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[18BS3CS02]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

MATHEMATICS FOR COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions
- **ii.** Questions 1 and 2 are compulsory.
- **iii.** Missing data may be suitably assumed.

1. a. i. State and prove addition principle for two sets. **07**

ii. A computer company requires 30 programmers to handle system programming jobs and 40 programmers for applications programming. If the company appoints 55 programmmers to carry out these jobs, how many of these perform jobs of both types? How many handle only system programming jobs? How many handle only applications programming.

b. State whether the argument given is valid or not. If it is valid, identify the tautology. **07**

If I try hard and I have talent, then I will become musician.
If I become a musician, then I will be happy.

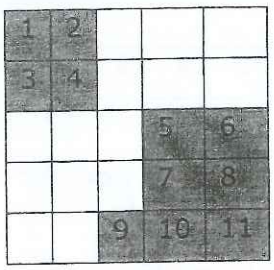
If I will not be happy, then I did not try hard or I do not have talent.

2. a. i. Let A= $\{1,2,3,4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm. **08**

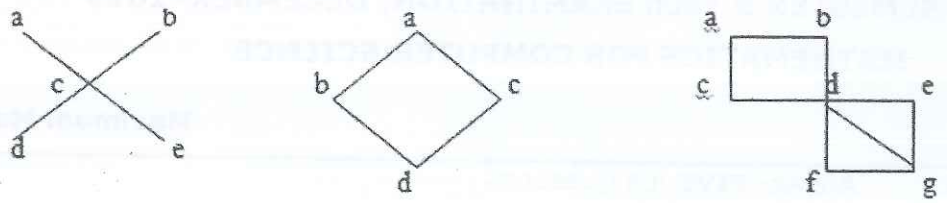
$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ii. Solve the recurrence relation $C_n = 3C_{n-1} - 2C_{n-2}$ with initial conditions $C_1 = 5$ and $C_2 = 3$.

b. Find the rook polynomial for the board shown below (shaded part) **06**



3. a. Determine whether the graph shown is a Euler path and circuit, Hamiltonian path and circuit. **06**

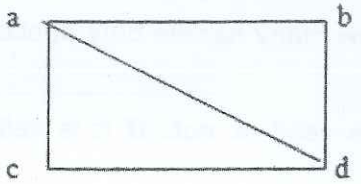


b. i. Define the following terms **08**

i. Chromatic number of graph G

ii. Chromatic polynomial of graph G

ii. Find chromatic polynomial P_G for the given graph and use P_G to find $\chi(G)$



OR

4. a. Define the following terms with examples **08**

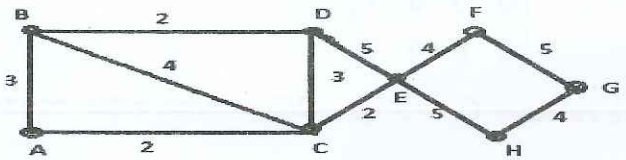
i. Bipartite graphs

ii. Planar graphs

iii. Euler circuits

iv. Graph coloring

b. Use Prim's algorithm to find minimal spanning tree for the graph given in the fig. Use vertex F as the initial vertex and list the edges in the order in which they are chosen. **06**



5. a. Find the correlation co-efficient and regression lines of y on x and x on y for the following data: **07**

x	1	6	5	10	3	2	4	9	7	8
y	6	4	9	8	1	2	3	10	5	7

b. Fit a parabola of second degree $y = a + bx + cx^2$ for the data **07**

x	0	1	2	3	4
y	1	1.8	1.3	2.5	2.3

OR

6. a. i. If A and B are events with $P(A)=1/2$, $P(A\cup B)=3/4$ and $P(\overline{B})=5/8$, find the following $P(A\cap B), P(\overline{A}\cap\overline{B}), P(\overline{A}\cup\overline{B}), P(\overline{A}\cap B)$ **08**

ii. A box 'A' contains 2 white and 4 black balls. Another box 'B' contains 5 white and 7 black balls. A ball is transformed from the box A to the box B. Then a ball is drawn from the box B. Find the probability that it is white.

b. In a bolt factory there are four machines A, B, C, D manufacturing respectively 20%, 15%, 25%, 40% of the total production. Out of these 5%, 4%, 3%, 2% are defective. If a bolt drawn at random from the production and is found to be defective. Find the probability that it was manufactured by A or D? **06**

7. a. In 800 families with 5 children each how many families would be expected to have (i) 3 boys (ii) 5 girls (iii)either 2 or 3 boys (iv) atmost 2 girls. By assuming probabilities for boys and girls to be equal. **07**

b. In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10, use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 10000 packets. **07**

OR

8. a. Suppose X and Y are independent random variables with the following distribution, find the joint distribution of X, Y also verify that $Cov(X,Y)=0$ **07**

x_i	1	2
$f(x_i)$	0.7	0.3

y_j	-2	5	8
$g(y_j)$	0.3	0.5	0.2

b. The kilometer run (in thousands of k.ms) without any sort of problem in respect of a certain vehicle is a random variable having p.d.f **07**

$$f(x)=\begin{cases}\frac{1}{40}e^{-x/40}, & x\geq 0\\ 0 & x\leq 0\end{cases}$$ Find the probability that the vehicle is trouble free (i) atleast for 25000 k.ms ii) atmost for 25000 k.ms (iii) between 16000 to 32000 k.ms